PUNJAB ENGINEERING COLLEGE (DEEMED TO BE UNIVERSITY), CHANDIGARH

MINOR PROJECT SYNOPSIS

OPTIMIZATION-BASED RESOURCE ALLOCATION FOR PUBLIC BUS FLEET SCHEDULING UNDER DEMAND VARIABILITY

(A case-study orientation using the Chandigarh Transport Undertaking)

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY in MATHEMATICS AND COMPUTING

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1. Introduction

Urban public transport is the backbone of mobility in rapidly growing Indian cities, and fixed-route buses are its most accessible and affordable mode. Transport undertakings must distribute a finite fleet over a network of routes whose demand varies across space and time. When scheduled departures do not follow passenger demand, the network suffers two failures at once: overcrowding and long waits on busy corridors, and under-used capacity on quiet ones.

Mathematically, this is a constrained multi-objective resource-allocation problem: a limited pool of buses must be allocated across routes and time periods to balance conflicting objectives. More departures reduce waiting and crowding but increase operating distance and cost; fewer departures save cost but hurt passengers. A useful framework must expose these trade-offs rather than produce a single schedule.

This project develops such a framework for bus fleet allocation and frequency setting under uncertain demand, with a case-study orientation on the Chandigarh Transport Undertaking (CTU), which runs local and tri-city services in Chandigarh, Mohali and Panchkula. It uses CTU's published timetable (routes, headways, route lengths and allocated buses) and is formulated generically, so that ridership data from any operator can be plugged in.

2. Problem Statement

Given a fixed number of buses, a network of fixed routes with known cycle times and published headway ranges, and passenger demand that varies by route and time period, determine for every route and period (i) a headway chosen from a discrete set of practical headways and (ii) the number of buses needed to run it. The schedule should minimise expected passenger waiting and overcrowding while keeping operating distance (vehicle-km) within budget, and must respect the fleet limit in every period, the buses required to sustain each frequency, the published frequency bounds and limits on how far allocations may change.

Demand varies from day to day and no ridership data are public, so demand is modelled as a triangular fuzzy number per route and period. The project tests whether a deterministic model evaluated against demand variation is sufficient, or whether the uncertainty should enter the constraints explicitly through α-cuts.

3. Mathematical Modeling Framework

To ground the resource allocation problem in a rigorous mathematical format, we establish a multi-objective optimization model. The formulation represents the transit network as a set of discrete routes and planning periods, and defines parameters and decision variables that can be calibrated with operational datasets. Every objective and constraint below is linear in the decision variables, so the whole model is a mixed-integer linear program (MILP) that standard solvers can handle.

3.1 Sets, Parameters and Decision Variables

The model uses the following set definitions:

- R: set of all bus routes in the network, indexed by r (here, the 68 CTU routes).
- T: set of planning time periods, indexed by t (here, $t = 1$ for AM 07–10, $t = 2$ for Midday 10–16 and $t = 3$ for PM 16–19).
- K_{rt}: set of admissible headways (in minutes) for route r in period t, indexed by k. It is the menu $\{10, 12, 15, 20, 25, 30, 40, 45, 60, 90\}$ together with the route's published headways, restricted to the route's published range.

The operational parameters and decision variables are summarised in the tables below.

Parameter symbol	Operational meaning and description
$B,\ B_{\text{eff}}$	Buses allocated to the operator's local and tri-city services (389) and buses available for service after spares, $B_{\text{eff}} = \lfloor (1-s)B \rfloor$
C	Effective passenger capacity of a single bus (seated capacity plus permitted standees)
$\tilde{D}_{rt} = (L, M, U)$	Triangular fuzzy hourly passenger demand on route r in period t: low, most plausible and high values (passengers per hour)
τ_r	Cycle time of route r in hours: running time for a full cycle plus layover and turnaround buffers
$d_r,\ d_r^{\text{cyc}}$	Published route length (one-way for non-loop routes) and distance covered per bus cycle ($2d_r$, or d_r for loops), in km
H_{rt}	Hours of service of route r inside period t
$f_r^{\min},\ f_r^{\max}$	Minimum and maximum published service frequencies (departures per hour) $= 60/$longest and shortest published headway
x_r^0	Current (baseline) number of buses on route r

Decision variable	Operational scope and optimization role
$y_{rtk} \in \{0, 1\}$	Binary variable equal to 1 if route r runs headway h_k, $k \in K_{rt}$, in period t
$f_{rt} = \sum_k (60/h_k)\, y_{rtk}$	Resulting service frequency (departures per hour); a linear expression in y_{rtk}, not a separate variable
$x_{rt} \in \mathbb{Z}_{\geq 0}$	Integer variable giving the number of buses allocated to route r during period t
$o_{rt} \geq 0$	Auxiliary variable measuring unaccommodated overcrowding (passengers per hour above capacity) on route r in period t

3.2 Model Objectives

The optimization seeks to minimise three conflicting objectives:

$$\min Z_1 = \sum_{r \in R} \sum_{t \in T} M_{rt} \sum_{k \in K_{rt}} \frac{h_k}{2} y_{rtk} \tag{1}$$

Objective Z_1 models the expected passenger waiting time across the network. Assuming random passenger arrivals, the average wait is half the headway, $h_k/2$ minutes, so running headway h_k on route r in period t costs $M_{rt}h_k/2$ passenger-minutes per hour. Weighting the wait by the most plausible demand M_{rt} prioritises high-demand routes, so a small improvement on a busy corridor moves the objective far more than the same improvement on a quiet one. Writing the wait through the binary headway choice y_{rtk}, rather than as $D \cdot 30/f$, keeps Z_1 linear; discrete frequency sets are an established way to do this.

$$\min Z_2 = \sum_{r \in R} \sum_{t \in T} d_r^{\text{cyc}} H_{rt} f_{rt} \tag{2}$$

Objective Z_2 represents the total operating effort in vehicle-km per day, which serves as a proxy for operating cost. Each departure covers d_r^{cyc} km, route r runs f_{rt} departures per hour for H_{rt} hours, and the product is the distance run. Minimising it controls fuel consumption, tyre wear and vehicle depreciation, and balances the passenger-side improvements against the operator's budget.

$$\min Z_3 = \sum_{r \in R} \sum_{t \in T} o_{rt} H_{rt} \tag{3}$$

Objective Z_3 measures the total network overcrowding, counted as passengers above capacity per day. Minimising the overflow variables o_{rt} over all routes and periods pushes capacity towards the busiest corridors and times, reducing the situations in which passengers are left behind by packed buses.

The objectives conflict: more departures lower Z_1 and Z_3 but raise Z_2, and fewer departures do the opposite. The model therefore exposes the trade-off instead of returning one schedule (Section 7).

3.3 Operational Constraints

The model enforces the following linear constraints on headway choice, fleet, capacity and service limits, for every route $r \in R$ and period $t \in T$ unless stated otherwise:

$$\sum_{k \in K_{rt}} y_{rtk} = 1, \qquad f_{rt} = \sum_{k \in K_{rt}} \frac{60}{h_k} y_{rtk} \tag{4}$$

Constraint (4) selects exactly one headway from the menu for each active route-period and defines the resulting frequency. A discrete menu mirrors how timetables are actually published, and because f_{rt} is linear in y_{rtk}, every constraint that uses it stays linear.

$$\sum_{r \in R} x_{rt} \leq B_{\text{eff}} \qquad \forall t \in T \tag{5}$$

Constraint (5) ensures that the total number of buses assigned across all routes in any period t does not exceed the buses available for service, B_{eff}, after a fraction s is set aside as spares. This is the fixed resource barrier of the transport undertaking.

$$\tau_r f_{rt} \leq x_{rt} \tag{6}$$

Constraint (6) is the frequency-fleet relationship: a route can only run departures as often as its allocated buses allow. A single bus completes one cycle every τ_r hours, so it supplies at most $1/\tau_r$ departures per hour. Thus x_{rt} buses can sustain at most x_{rt}/τ_r departures per hour, which gives the inequality.

$$o_{rt} \geq D_{rt}^{\alpha} - C f_{rt}, \qquad o_{rt} \geq 0 \tag{7}$$

Constraint (7) defines the overcrowding variable o_{rt}. Hourly capacity is $C \cdot f_{rt}$, because it depends on how often buses depart, not on how many are allocated. If the design demand D_{rt}^{α} (Section 3.4) exceeds this capacity, o_{rt} takes the positive difference, representing unaccommodated demand; if capacity is sufficient, non-negativity forces $o_{rt} = 0$.

$$f_r^{\min} \leq f_{rt} \leq f_r^{\max} \tag{8}$$

Constraint (8) imposes the operational frequency bounds. The frequency must remain within the published range: the lower bound secures equity and minimum service commitments, and the upper bound prevents congestion and redundant scheduling. A maximum headway of 60 minutes in the peaks and 90 minutes at midday is treated as an additional equity goal.

$$|x_{rt} - x_r^0| \leq \max\bigl(1,\ \min(3,\ 0.3\,x_r^0)\bigr) \tag{9}$$

Constraint (9) limits how far an allocation may move from the current one x_r^0: by at most 30% of the current buses, capped at three buses, but never less than one bus. This keeps the recommended schedules realistic for depots and crews.

$$y_{rtk} \in \{0,1\}, \qquad x_{rt} \in \mathbb{Z}_{\geq 0}, \qquad o_{rt} \geq 0 \tag{10}$$

Constraint (10) defines the variable domains. The headway choice is binary, and the bus allocation x_{rt} must be an integer because fractional buses are physically meaningless, whereas the overflow o_{rt} is continuous. This structure follows the classical fleet-constrained frequency allocation of Furth and Wilson.

3.4 Modelling Demand Uncertainty: Optimization under Vagueness

In real transit operations, passenger demand is rarely a fixed number. Ridership varies with weather, traffic patterns, seasonal holidays and socio-economic factors, and no ridership data are public for CTU. To capture this uncertainty without the heavy data requirements of full probability distributions, the project uses fuzzy numbers.

Hourly demand is represented as a Triangular Fuzzy Number (TFN) $\tilde{D}_{rt} = (L_{rt}, M_{rt}, U_{rt})$, where L_{rt} is the low demand, M_{rt} the most plausible (modal) demand and U_{rt} the high demand, all in passengers per hour. For capacity planning, the high value U is the pessimistic case, because it is the load the buses must be able to carry.

A probabilistic chance constraint $P(\text{capacity} \geq \text{demand}) \geq 1-\alpha$ would express that capacity covers demand with high probability, but it requires a probability distribution, which three fuzzy values do not provide. The model therefore uses the possibilistic α-level constraint: at level $\alpha \in [0,1]$, capacity must cover the upper end of the α-cut of the demand,

$$C f_{rt} \geq D_{rt}^{\alpha} = U_{rt} - \alpha\,(U_{rt} - M_{rt}) \tag{11}$$

This is a deterministic linear constraint, so standard MILP solvers can process the model while still respecting a safety margin. The level α controls how conservative the plan is. With $\alpha = 1$ the plan covers the most plausible demand M; with $\alpha = 0$ it covers the high value U; values in between interpolate, so lowering α builds a larger capacity buffer. The waiting objective uses M (the expected passenger experience), whereas the capacity constraint uses D^{α} (a safety requirement). Constraint (11) is the form used in (7).

4. Proposed System Methodology

The workflow is a reproducible pipeline from data to policy evaluation (Figure 1). Each stage produces checked outputs, and the pipeline is re-run as better data arrive. Stages 1 to 6 are the linear-model pipeline; stage 7 is the next step, covering the metaheuristic and simulation extensions.

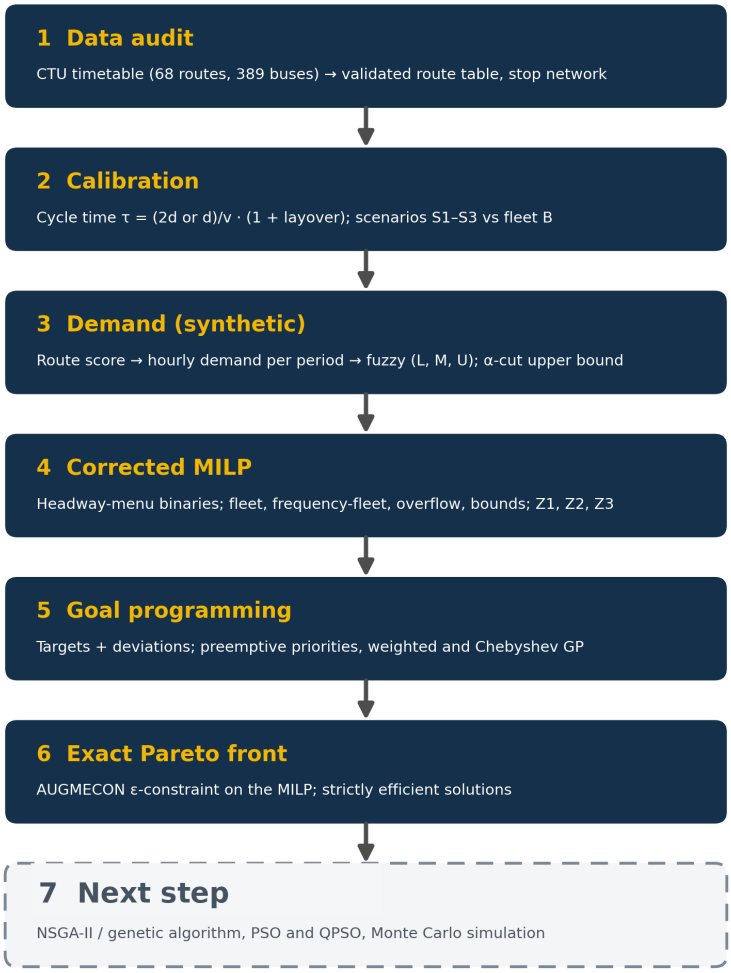


Figure 1: Optimization and system methodology

5. International Benchmarking and Best Practices

Transport for London plans bus service levels against published planning guidelines, and Singapore's Bus Service Reliability Framework rewards or penalises operators according to Excess Wait Time (EWT), the extra waiting caused by irregular service. We adopt their principles rather than their numbers: allocation is driven by demand at the busiest times, and reliability is measured by $\mathrm{EWT} = W^{\mathrm{actual}} - W^{\mathrm{sched}}$, with $W^{\mathrm{sched}} = h/2$ for a scheduled headway h. In the planned simulation extension (Section 7), irregularity will be represented, until arrival (AVL) data are available, by headways with a coefficient of variation CV, giving an expected wait of $(h/2)(1+\mathrm{CV}^2)$ and $\mathrm{EWT} = (h/2)\,\mathrm{CV}^2$. The model is a repeatable decision-support engine that can be re-run as demand changes, in the spirit of continuous network review.

6. Data Ingestion and Baseline Construction

The published CTU timetable is the primary input. Parsing and validating it gives 68 routes, 389 allocated buses, 1,662 km of route length and 649 named stops, with every route checked against

the timetable PDF. Cycle times are calibrated so that the published frequencies are consistent with the allocated fleet; the evidence favours reading the published length as one-way, which CTU should confirm.

For passenger demand, official ridership data (ticket-machine records or boarding counts) have been requested through CTU. Until they are available, demand is constructed synthetically from documented structural features (route length, service to major hubs such as PGI and the ISBTs, and number of hub stops) and scaled to a network-wide daily total. All results based on it are labelled synthetic and illustrative. A manual boarding count at 5 to 10 hub stops is planned to validate and replace the synthetic values, with L, M and U estimated as the 10th, 50th and 90th percentiles of observed demand.

7. Proposed Solution Methods

The solution methods are the linear (exact) methods; heuristic and simulation-based methods are future work (item 4).

1. **Exact optimization (MILP).** The corrected model is a mixed-integer linear program. At the full CTU network size (191 active route-periods, 651 variables) it is solved to proven optimality by branch and bound with an open-source solver. Trade-offs are traced with the augmented ε-constraint method (AUGMECON), which returns only strictly Pareto-efficient schedules.
2. **Goal programming with priorities.** Each objective becomes a goal with a target; preemptive (priority-ordered), weighted and min-max goal programming generate passenger-first, operator-first, equity-first and balanced schedules.
3. **Fuzzy constraints.** Fuzzy demand enters through the crisp α-cut constraint (11), so the model remains an MILP and all of the above methods apply to it unchanged.
4. **Future work: metaheuristic search and simulation.** Future extensions will complement the exact linear methods with the following.
   - *NSGA-II (genetic algorithm).* NSGA-II will approximate the Pareto front in a single run. It will be benchmarked against the exact AUGMECON front and is the route to nonlinear or simulation-based extensions.
   - *Particle swarm optimization (PSO and QPSO).* Particle Swarm Optimization (PSO) and Quantum-behaved Particle Swarm Optimization (QPSO) will be investigated as alternative metaheuristics for large-scale or computationally demanding transit optimization problems. They will be evaluated against NSGA-II and the MILP baseline in terms of solution quality, convergence, computational time and robustness. For vehicle routing and scheduling extensions, adapted PSO/QPSO variants may also be explored to handle discrete route decisions and multiple operational objectives.
   - *Monte Carlo simulation.* The resulting schedules will be stress-tested by Monte Carlo simulation of demand and headway irregularity, including $\pm 10\%$ and $\pm 20\%$ demand changes.

8. Software Tools and Project Plan

Category	Tool and purpose
Programming	Python (pandas, NumPy, SciPy, Matplotlib, openpyxl) for data, analysis and figures
Optimization	PuLP with the HiGHS solver for the MILP, goal programming and the ε-constraint method
Metaheuristic (future work)	DEAP for NSGA-II
Fuzzy demand	Closed-form triangular fuzzy numbers and α-cuts (no separate library needed)
Testing	pytest for automated checks of data, model and solvers

9. Expected Outcomes

- **Mathematical framework.** A corrected, linear multi-objective model for bus fleet allocation and frequency setting under fuzzy demand.
- **Computational toolkit.** A reproducible Python codebase with exact MILP, goal programming and AUGMECON, with automated tests. NSGA-II, PSO/QPSO and Monte Carlo evaluation are planned extensions.
- **Baseline and gap analysis.** A validated CTU baseline that identifies route-periods where supply and demand are mismatched.
- **Allocation alternatives.** Pareto-efficient and priority-based schedules. The hypothesis to test with real demand data is that waiting and overcrowding can be reduced without additional buses.
- **Report and paper.** A final report and a research paper describing the method and the case study.

10. Conclusion

This synopsis presents a mathematically sound approach to bus fleet allocation under demand variability for the Chandigarh Transport Undertaking. The formulation balances passenger interests, including waiting time and crowding, with operator interests, such as vehicle-kilometres, through a linear multi-objective optimization model. Uncertain demand is represented using fuzzy α-cuts, while exact optimization (MILP with AUGMECON) and goal programming are employed to obtain and compare solutions under different priorities. Metaheuristic search (NSGA-II, PSO/QPSO) and Monte Carlo simulation are identified as future extensions. The approach provides a systematic framework for identifying efficient, practical and robust bus allocation schedules.

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